

- Inference on population proportion

- Single proportion

Simple random variable that follow a $\text{Binomial}(n, p)$ distribution

- You can construct confidence intervals or
- Conduct a hypothesis test

- Difference of proportion

Two simple random variables that follow $\text{Binomial}(n_1, p_1)$ and $\text{Binomial}(n_2, p_2)$ distributions

- You can construct confidence intervals or
- Conduct a hypothesis test

Did first-class passengers have better chances of surviving than third-class passengers on board the Titanic?

- Inference on population mean

- Single mean

Simple random variable that follows some distribution $D(\mu, \sigma)$

- You can construct confidence intervals or
- Conduct a hypothesis test

- Difference of mean

Two simple random variables that follow some distributions $D(\mu_1, \sigma_1)$ and $D(\mu_2, \sigma_2)$ distributions

- You can construct confidence intervals or
- Conduct a hypothesis test

Is there higher NO₂ levels in air categorized as hazardous than good air quality?

- Inference for simple linear regression

You can consider a simple linear regression given by the equation

$$Y = \beta_0 + \beta_1 X + \epsilon_i$$

- Construct a confidence interval for the slope, β_1
- Conduct hypothesis testing for β_1

Is there a linear relationship between the number of licensed drivers and the CO₂ emissions in a given state?